

Cloud Computing

1. What is Cloud Computing?

Professional Definition

Cloud Computing is the on-demand delivery of computing services such as servers, storage, databases, networking, software, analytics, and artificial intelligence over the internet, allowing users to access these resources without owning or maintaining physical infrastructure.

Cloud service providers like Microsoft Azure, Amazon Web Services (AWS), and Google Cloud Platform (GCP) manage the underlying infrastructure, while users pay only for the resources they consume.

Key Features

- On-demand resource provisioning
- Pay-as-you-go pricing
- Scalability and elasticity
- High availability
- Global accessibility
- Managed infrastructure

Example

Instead of purchasing and maintaining a physical server to host a website, an organization can deploy it on Microsoft Azure and pay only for the computing resources used.

2. What are the Advantages of Microsoft Azure?

Professional Definition

Microsoft Azure is Microsoft's cloud computing platform that provides a comprehensive collection of cloud services for building, deploying, and managing applications through Microsoft's global network of data centers.

Advantages

1. Scalability

Azure enables organizations to increase or decrease computing resources based on business requirements without purchasing additional hardware.

2. Cost Efficiency

Azure follows a **pay-as-you-go** pricing model, allowing businesses to pay only for the resources they consume.

3. High Availability

Azure provides built-in redundancy and disaster recovery capabilities, ensuring applications remain available even during hardware or regional failures.

4. Security and Compliance

Azure offers enterprise-grade security features such as:

- Identity and access management
 - Data encryption
 - Firewall protection
 - Compliance with international standards (ISO, GDPR, HIPAA, etc.)
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5. Global Infrastructure

Azure operates multiple data centers across different geographic regions, enabling organizations to deploy applications closer to users for lower latency and better performance.

6. Wide Range of Services

Azure provides numerous cloud services including:

- Virtual Machines
- Azure Storage
- Azure SQL Database
- Azure Data Factory
- Azure Databricks
- Azure Kubernetes Service

- Azure AI Services
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7. Easy Integration

Azure integrates seamlessly with Microsoft products such as:

- Windows Server
 - Active Directory
 - Microsoft 365
 - Visual Studio
 - GitHub
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3. What is Hosting in Azure?

Professional Definition

Hosting in Azure refers to deploying and running applications, websites, APIs, databases, or virtual machines on Microsoft Azure's cloud infrastructure, making them accessible over the internet.

Azure manages the underlying hardware, networking, storage, and security, while users deploy and manage their applications.

Types of Azure Hosting

- Azure Virtual Machines
- Azure App Service
- Azure Kubernetes Service (AKS)
- Azure Static Web Apps
- Azure Functions (Serverless)

Example

A company develops an e-commerce website and hosts it on **Azure App Service**, allowing customers worldwide to access the application through the internet.

4. What is IaaS, PaaS, and SaaS?

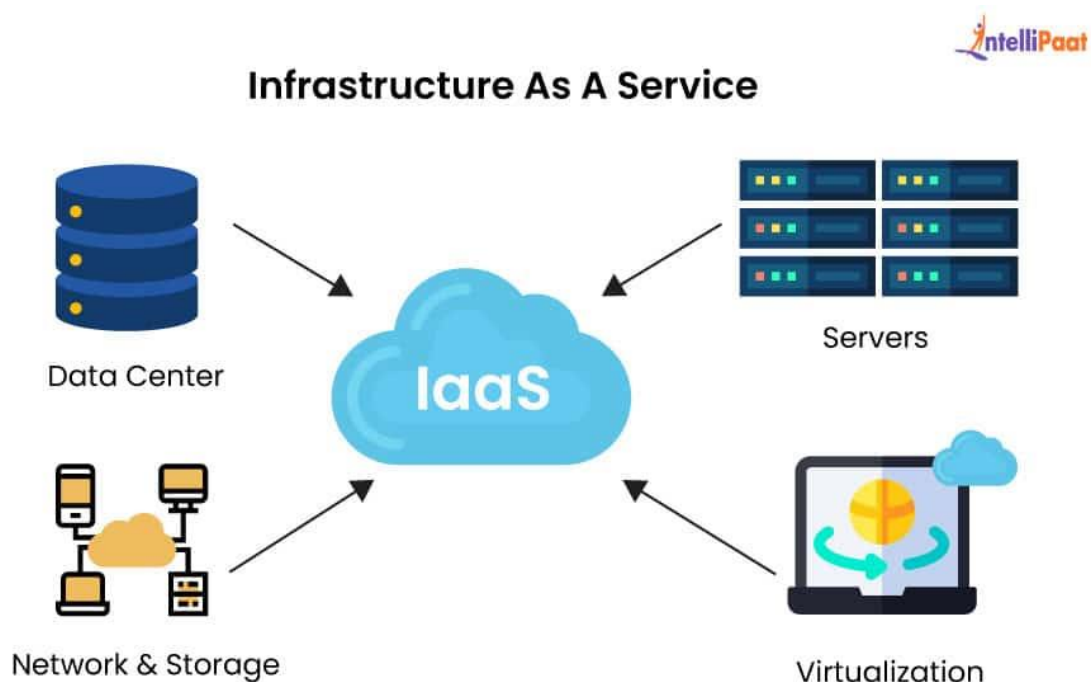
Cloud services are generally classified into three service models based on the level of responsibility shared between the cloud provider and the customer.

IaaS (Infrastructure as a Service)

Definition

Infrastructure as a Service (IaaS) provides virtualized computing resources such as virtual machines, storage, and networking over the internet.

The cloud provider manages the physical infrastructure, while the customer is responsible for managing the operating system, middleware, applications, and data.



Customer Manages

- Operating System
- Applications
- Runtime
- Data

Cloud Provider Manages

- Physical servers
- Storage

- Networking
- Virtualization

Azure Example

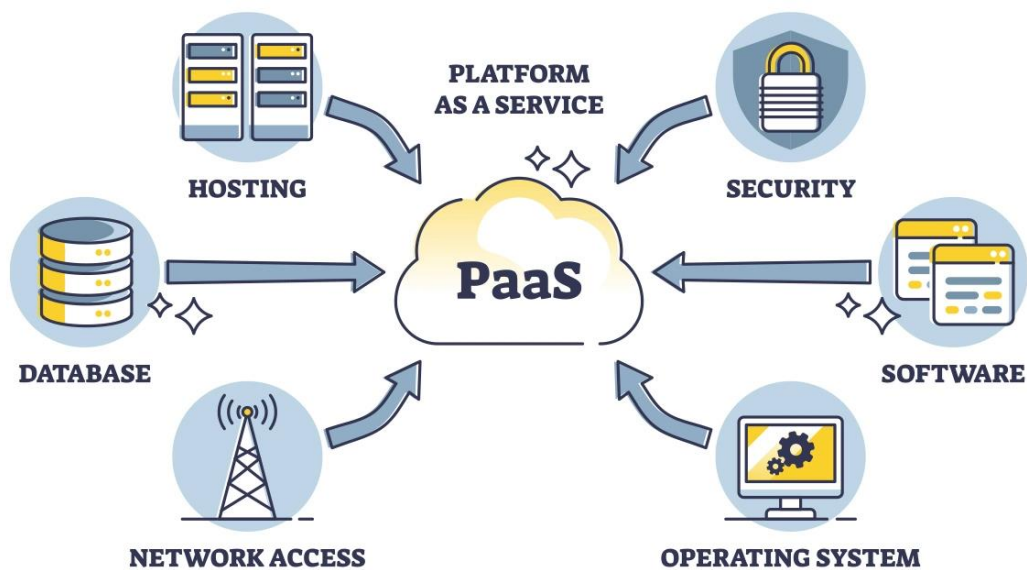
- Azure Virtual Machine
- Azure Virtual Network
- Azure Managed Disks

PaaS (Platform as a Service)

Definition

Platform as a Service (PaaS) provides a complete platform for developing, testing, deploying, and managing applications without managing the underlying infrastructure.

The cloud provider manages the servers, operating system, runtime, and middleware, allowing developers to focus only on writing application code.



Customer Manages

- Application code
- Data

Cloud Provider Manages

- Infrastructure

- Operating System
- Runtime
- Middleware
- Networking

Azure Example

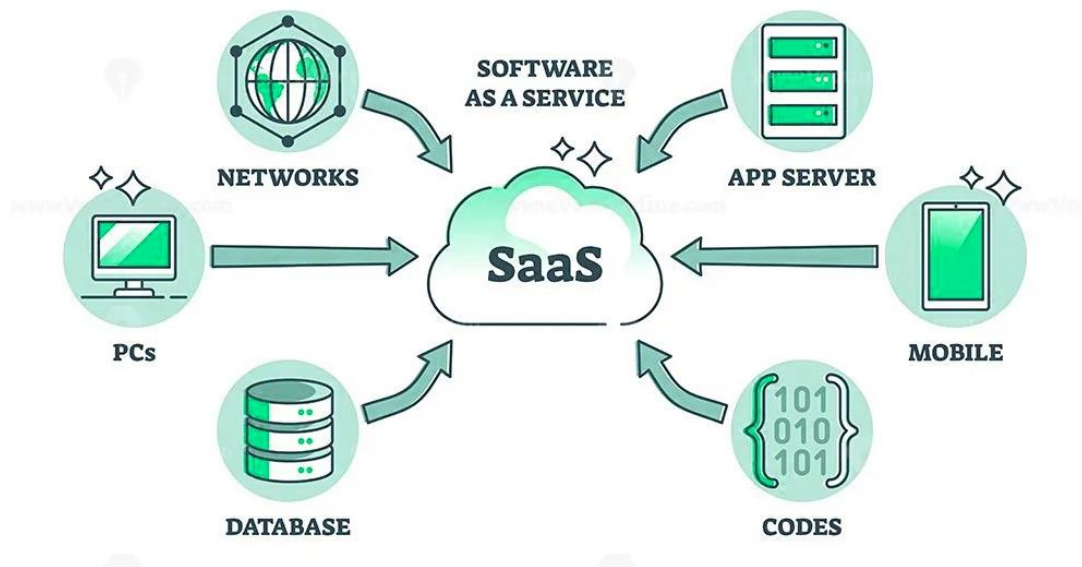
- Azure App Service
- Azure SQL Database
- Azure Functions

SaaS (Software as a Service)

Definition

Software as a Service (SaaS) delivers fully functional software applications over the internet. Users simply access the application through a web browser or client application without installing or maintaining the software.

The cloud provider manages the complete application and infrastructure.



Customer Manages

- User settings
- Application usage

Cloud Provider Manages

- Application
- Infrastructure
- Updates
- Security
- Maintenance

Examples

- Microsoft 365
- Gmail
- Salesforce
- Dropbox
- Zoom

Comparison of IaaS, PaaS, and SaaS

Feature	IaaS	PaaS	SaaS
Infrastructure Managed By	Cloud Provider	Cloud Provider	Cloud Provider
Operating System Managed By	Customer	Cloud Provider	Cloud Provider
Runtime Managed By	Customer	Cloud Provider	Cloud Provider
Application Managed By	Customer	Customer	Cloud Provider
User Responsibility	Infrastructure configuration and application management	Application development and deployment	Using the software
Azure Example	Azure Virtual Machine	Azure App Service	Microsoft 365